

**General
Structure**

Garcia-Suarez (2022) JMPS $A \in \mathfrak{sl}_2(\mathbb{R})$

$$\frac{d}{dx} \begin{bmatrix} f \\ f' \end{bmatrix} = \begin{bmatrix} 0 & A_{12} \\ A_{21} & 0 \end{bmatrix} \begin{bmatrix} f \\ f' \end{bmatrix}$$

$$\rightarrow f(x_{k+1}) = \overbrace{\exp(A_k l)}^{T_k(x_{k+1}, x_k) \in SL_2(\mathbb{R})} f(x_k)$$



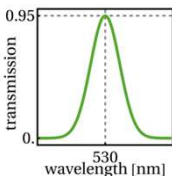
Acoustics

*anti-sonar
coating
(anechoic tile)*



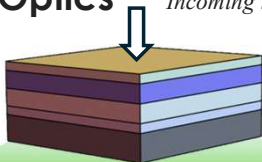
$$A = 1 - |R| - |T|$$

*Absorption
efficiency ≈ 1*



Optics

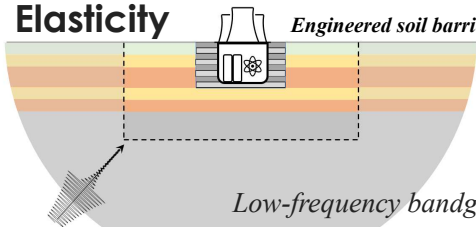
Incoming light



color filter

Elasticity

Engineered soil barrier



Low-frequency bandgaps